

Linux Notes

Complete Guide: Theory, Types and Commands

Based on class notes dated 28/09, with corrections and additions

Part	Contents
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Part 2	Types: software, OS, kernels, users, files, shells, distros, and more
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Part 1: Theory

1. Software (S/W)

A set of instructions (programs) that tells hardware what to do. Hardware is the physical part; software is the logic that runs on it.

2. Operating System (OS)

System software that acts as a **bridge between the user and the hardware**. It manages the CPU, memory, storage and devices.

- Process management (what runs, and when)
- Memory management
- File management
- Device management
- Security and user access
- Provides the user interface (GUI or CLI)

Properties of a good OS

- Efficient use of resources
- Security and protection
- Stability and reliability
- Portability
- Multitasking and multi-user support
- Easy for the user to work with

3. History of Linux

- **Unix** was created at Bell Labs in the 1970s.
- **GNU** project (1983, Richard Stallman) aimed to build a free Unix-like OS. It had tools but no kernel.
- **Linux kernel** was written by **Linus Torvalds** in 1991.
- GNU tools plus the Linux kernel make a complete OS, sometimes called GNU/Linux.

4. GNU and GPL

- **GNU** = "GNU's Not Unix", a project for free software.
- **GPL** = General Public License. Anyone can use, study, modify and share the software, but modified versions must stay open under the same license.
- This is why Linux is free and open source.

5. Kernel

The **core of the OS**. It loads first at boot and stays in memory.

- Controls the CPU, RAM and devices
- Manages processes and scheduling
- Handles system calls from programs
- Enforces security and permissions
- Linux uses a **monolithic kernel**: main services run inside the kernel.

6. Kernel Space vs User Space

Kernel space	User space
Runs the kernel	Runs apps, shell, user programs
Full access to hardware and memory	Restricted access
A crash can bring down the whole system	A crash affects only that program
Privileged mode	Unprivileged mode

User-space programs request services from the kernel through **system calls**.

7. Shell and CLI

- **CLI** (command-line interface): you type commands instead of clicking.
- **Shell**: the program that reads your commands and passes them to the kernel.
- Flow: You -> Shell -> Kernel -> Hardware
- **Bash** (Bourne Again Shell) is the default on Ubuntu.
- **Terminal**: the window where you use the shell.
- Prompt: **#** = root user, **\$** = general user.

8. User Types

- **Root/Admin**: full control
- **General user**: limited permissions
- **Service user**: used by programs and services (e.g. a Google account), not by people

9. File Permissions

Shown as **rwxr-xr--** in three groups: **owner, group, others**. **r** = read (4), **w** = write (2), **x** = execute (1).
Example: **chmod 755 file** gives the owner rwx, and group and others r-x.

10. Boot Process

- **BIOS/UEFI** checks the hardware
- **GRUB** (boot loader) loads the kernel
- **Kernel** starts and mounts the filesystem
- **systemd** (init) starts services
- Login prompt or desktop appears

11. Linux Filesystem Idea

- Everything is a **file**, even devices.
- Single tree starting at **/** (root).
- Case-sensitive: **File.txt** and **file.txt** are different.

12. Important Directories

Path	Purpose
/	Root of the whole filesystem
/home	User home directories
/bin	Essential binaries (commands)
/usr	Installed programs and libraries
/etc	Configuration files

Path	Purpose
/tmp	Temporary files
/var/log	Log files
/proc	Live system and process info
/dev	Device files
/boot	Boot loader and kernel files
/root	Home directory of the root user

Part 2: Types

1. Types of Software

Type	Examples
System software	OS, device drivers, firmware, utilities
Application software	Browser, MS Office, VLC, games
Programming software	GCC, Python interpreter, VS Code
Middleware	Database connectors, message brokers

By license: Open source (Linux, Firefox); Proprietary (Windows, MS Office); Freeware (free, closed source); Shareware (free trial, then pay); Public domain (no restrictions).

2. Types of Operating Systems

Type	Meaning	Example
Batch	Jobs grouped and run without interaction	Early IBM mainframes
Time-sharing	CPU time split among users	Unix
Multiprogramming	Many programs kept in memory	Most modern OS
Multitasking	Several tasks run seemingly together	Windows, Linux
Multi-user	Many users at once	Linux, Unix
Real-time	Strict response time	VxWorks, FreeRTOS
Distributed	Networked machines act as one	LOCUS
Network OS	Manages network resources	Windows Server
Embedded	Built into devices	Router firmware
Mobile	Phones and tablets	Android, iOS

By interface: CLI (typed commands) and GUI (windows and icons). **By architecture:** 32-bit and 64-bit.

3. Types of Kernels

- **Monolithic:** all services in one big kernel, fast. Example: Linux.
- **Microkernel:** only core functions in kernel, rest in user space, more stable. Example: QNX, MINIX.
- **Hybrid:** mix of both. Example: Windows NT, macOS.

4. Types of Users in Linux

- **Root (superuser):** UID 0, full control
- **Regular user:** UID usually 1000 and above
- **System/service users:** UID below 1000, run daemons (e.g. `www-data`)
- **Groups:** collections of users sharing permissions

5. Types of Files (first character of `ls -l`)

Symbol	Type
-	Regular file
d	Directory
l	Symbolic link

Symbol	Type
c	Character device
b	Block device
s	Socket
p	Named pipe

6. Types of Shells

- **sh**: original Bourne shell
- **bash**: Bourne Again Shell (Ubuntu default)
- **zsh**: modern, customizable (macOS default)
- **ksh**: Korn shell; **csch/tcsh**: C shell; **fish**: user-friendly
- Check yours with **echo \$SHELL**; list all with **cat /etc/shells**

7. Types of Permissions

Permission	On a file	On a directory
r (4)	read contents	list contents
w (2)	modify	create/delete files inside
x (1)	run as program	enter with cd

Who: **u** owner, **g** group, **o** others, **a** all.

8. Types of Linux Distros

Family	Distros	Package manager
Debian	Debian, Ubuntu, Mint, Kali	apt , .deb
Red Hat	RHEL, CentOS, Fedora, Rocky	dnf/yum , .rpm
Arch	Arch, Manjaro	pacman
SUSE	openSUSE, SLES	zypper

9. Types of Commands

- **Internal (built-in)**: part of the shell (**cd**, **echo**, **pwd**)
- **External**: separate programs (**ls**, **cat**, **cp**)
- **Aliases** and **functions**: user-defined. Check with **type command**.

10. Types of Redirection

- **>** overwrite output to a file
- **>>** append output to a file
- **<** take input from a file
- **2>** redirect errors
- **|** pipe output of one command into another (**ls | wc -l**)

11. Types of Processes

- **Foreground**: runs in the terminal
- **Background**: started with **&**

- **Daemon:** background service (sshd)
- **Zombie:** finished but not cleaned up
- **Orphan:** parent has ended

12. Types of Links

- **Hard link:** another name for the same data (`ln file link`)
- **Soft/symbolic link:** shortcut to a path (`ln -s file link`)

13. Types of Memory

- **RAM:** main working memory
- **Swap:** disk space used when RAM is full
- **Cache:** fast memory near the CPU
- **Virtual memory:** RAM plus swap, seen by programs as one space

Part 3: Commands From Your Notes

Navigation and Directories

Command	Description
<code>pwd</code>	Present working directory (where you are)
<code>ls</code>	List the files
<code>cd</code>	Change directory (alone, goes to home)
<code>cd folder</code>	Go into that folder
<code>cd ..</code>	Go back one level
<code>cd ~</code>	Go to home directory
<code>cd /</code>	Go to root of the filesystem
<code>mkdir name</code>	Create a new directory
<code>mkdir -p AA/BB/CC</code>	Create nested sub-directories
<code>sudo mkdir -p /root1/{a,b}</code>	Create a and b inside /root1
<code>sudo mkdir -p parent/{a,b}</code>	Create a and b inside parent

ls Options

Command	Description
<code>ls -a</code>	Show hidden files (names start with <code>.</code>)
<code>ls -lt</code>	Long format, sorted by time (your notes say <code>-it</code>)
<code>ls -S</code>	Sort by size, largest first
<code>ls ~</code>	List home directory
<code>ls *</code>	List files and contents of subfolders
<code>ls -R</code>	List recursively
<code>ls -r</code>	Reverse order
<code>ls -d */</code>	List only directories

OS and Kernel Info

Command	Description
<code>uname</code>	Kernel name (Linux)
<code>uname -a</code>	All info related to OS and kernel
<code>uname -s</code>	Kernel name
<code>uname -v</code>	Kernel version (build info)
<code>uname -m</code>	Machine architecture
<code>uname -n</code>	Host name
<code>uname -r</code>	Kernel release
<code>cat /etc/os-release</code>	Print OS metadata

Host, Hardware and Memory

Command	Description
<code>hostnamectl</code>	All info about host
<code>hostname</code>	Gives host name
<code>hostname -I</code>	IP address
<code>hostname -f</code>	Host name with domain name
<code>sudo hostnamectl set-hostname NEW</code>	Change hostname
<code>lscpu</code>	Description of CPU architecture
<code>cat /proc/cpuinfo</code>	Detailed info for every logical processor
<code>nproc</code>	Number of processors
<code>sudo dmidecode</code>	BIOS/hardware info
<code>free</code>	Physical RAM and swap memory
<code>vmstat -s</code>	Virtual memory statistics

Disk

Command	Description
<code>df</code>	Available and used space
<code>df -h</code>	Human-readable output (GB, MB instead of bytes)
<code>lsblk</code>	List block devices: disks and partitions
<code>sudo fdisk -l</code>	Info about disks and their partitions

Files and Text

Command	Description
<code>cat file</code>	Short for concatenate; combine and view files
<code>echo "xyz"</code>	Print text
<code>echo "xyz" >> one.txt</code>	Append text to a file
<code>echo "this is me" > two.txt</code>	Write text to a file (overwrites)
<code>cat two.txt</code>	Show contents of two.txt

Network and Status

Command	Description
<code>ip a / ip addr</code>	Show network interfaces and IPs
<code>ifconfig</code>	Older network configuration tool
<code>uptime</code>	Time since boot and load average

Users and Processes

Command	Description
<code>top</code>	Live view of running processes
<code>htop</code>	Friendlier version of top
<code>last</code>	Login history

Command	Description
<code>whoami</code>	Current username
<code>echo \$USER</code>	Current username from variable
<code>who</code>	Users logged in now
<code>who -r</code>	Current runlevel
<code>whatis cmd</code>	One-line description (your notes say <code>what</code>)

Date and Calendar

Command	Description
<code>date</code>	Full date and time
<code>date +%T</code>	Time (HH:MM:SS)
<code>date +%A</code>	Weekday name
<code>date +%B</code>	Month name
<code>date +%d</code>	Day of month
<code>date +%Y</code>	Year
<code>date +%m</code>	Month number
<code>date +%H</code>	Hour
<code>date +%M</code>	Minute
<code>date +%S</code>	Second
<code>date "+%d-%B-%Y"</code>	Custom format
<code>cal</code>	Current month
<code>ncal</code>	Vertical calendar
<code>cal -y</code>	Full year
<code>cal -3</code>	Previous, current, next month
<code>cal 2027</code>	Calendar of 2027

Part 4: Extra Commands (Complete Reference)

Files and Directories

Command	Description
<code>rmdir d</code>	Remove empty directory
<code>touch f</code>	Create empty file / update timestamp
<code>cp a b, cp -r d1 d2</code>	Copy file / directory
<code>mv a b</code>	Move or rename
<code>rm f, rm -r d</code>	Delete file / directory
<code>ln -s target link</code>	Symbolic link
<code>find / -name x</code>	Search by name
<code>locate x</code>	Fast search using index
<code>which cmd</code>	Path of a command
<code>file f</code>	Detect file type
<code>stat f</code>	Detailed file info

Viewing and Searching Text

Command	Description
<code>less f, more f</code>	Scroll through a file (q to quit)
<code>head -n 5 f</code>	First 5 lines
<code>tail -n 5 f</code>	Last 5 lines
<code>tail -f f</code>	Follow file live (logs)
<code>wc -l f</code>	Count lines
<code>nano f, vi f</code>	Edit a file
<code>diff a b</code>	Compare files
<code>grep "txt" f</code>	Find text in file (-i ignore case, -r recursive, -n line numbers, -v invert)
<code>sort f, uniq</code>	Sort / remove duplicate lines
<code>cut -d, -f1 f</code>	Extract a column

Permissions and Users

Command	Description
<code>chmod 755 f</code>	Set permissions (numeric)
<code>chmod u+x f</code>	Add execute for owner
<code>chown user:group f</code>	Change owner and group
<code>id</code>	UID and GID of current user
<code>sudo cmd</code>	Run as root
<code>su - user</code>	Switch user
<code>sudo adduser name</code>	Create user
<code>sudo passwd name</code>	Set password

Command	Description
<code>sudo userdel -r name</code>	Delete user and home
<code>sudo usermod -aG grp name</code>	Add user to group
<code>groups</code>	List groups

System, Disk and Processes

Command	Description
<code>lsusb, lspci</code>	List USB / PCI devices
<code>du -sh dir</code>	Size of a directory
<code>mount, umount</code>	Attach / detach filesystems
<code>ps aux</code>	All running processes
<code>kill PID, kill -9 PID</code>	Stop / force stop a process
<code>killall name</code>	Kill by name
<code>pgrep name</code>	Find PID by name
<code>cmd &</code>	Run in background
<code>jobs, fg, bg</code>	Manage jobs
<code>nohup cmd &</code>	Keep running after logout

Networking

Command	Description
<code>ping host</code>	Test connectivity
<code>curl url</code>	Fetch a URL
<code>wget url</code>	Download a file
<code>ss -tuln</code>	Show open ports
<code>traceroute host</code>	Trace route
<code>nslookup, dig</code>	DNS lookup
<code>ssh user@host</code>	Remote login
<code>scp f user@host:/path</code>	Copy over SSH
<code>sudo ufw enable</code>	Enable firewall (Ubuntu)

Packages, Archives, Services

Command	Description
<code>sudo apt update</code>	Refresh package list
<code>sudo apt upgrade</code>	Upgrade packages
<code>sudo apt install pkg</code>	Install a package
<code>sudo apt remove pkg</code>	Remove a package
<code>tar -czvf a.tar.gz dir</code>	Create compressed archive
<code>tar -xzvf a.tar.gz</code>	Extract archive
<code>zip -r a.zip dir, unzip a.zip</code>	Zip / unzip

Command	Description
<code>systemctl status svc</code>	Check a service
<code>sudo systemctl start/stop/restart svc</code>	Control a service
<code>sudo systemctl enable svc</code>	Start at boot
<code>journalctl -xe</code>	Recent logs and errors
<code>dmesg</code>	Kernel messages

Shell Tips and Power

Command	Description
<code>man cmd, cmd --help</code>	Help
<code>history, !!</code>	Past commands / repeat last
<code>clear</code> or <code>Ctrl+L</code>	Clear screen
<code>alias ll='ls -l'</code>	Create a shortcut
<code>export VAR=val</code>	Set environment variable
<code>env, printenv</code>	List variables
<code>Ctrl+C / Ctrl+Z / Ctrl+D</code>	Stop / suspend / exit
<code>Tab</code>	Autocomplete
<code>sudo reboot</code>	Restart
<code>sudo shutdown now</code>	Power off

Part 5: Quick Revision

Corrections to remember

- `cd` changes directory. `pwd` shows current directory.
- `/etc` holds configuration files. Executables live in `/bin` and `/usr/bin`.
- `/tmp` is for temporary files. Logs go in `/var/log`.
- `>` overwrites a file; `>>` appends.
- `uname -r` is the kernel release; `uname -s` is the kernel name.
- `#` is the root prompt, `$` is a general user. `%` is a zsh prompt.
- `ls -lt` sorts by time (not `-it`).

Practice questions

- What is the difference between kernel space and user space?
- Which command shows disk space in human-readable form?
- What is the difference between `>` and `>>`?
- How do you create `AA/BB/CC` in one command?
- Which command shows the kernel release?
- What does GPL allow?
- Name the steps of the Linux boot process.
- What is the difference between a hard link and a soft link?